Linnæus University
Sweden

Report

Wind energy in Canada

Author: Sandrine DANIEL
Supervisor:
Examiner:
University: Linnaeus University
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Linnæus University
Sweden

Source and country

Wind energy is a renewable energy source that is produced by using the kinetic energy of wind to move the turbine blades. The kinetic energy is then converted into rotational energy, which is transferred into a generator, therefore producing electrical energy[1]. This source of energy is now the most efficient source of clean energy[2].

The history of wind energy in Canada dates back over two centuries. In fact, it started with basic applications such as water pumping in the agricultural sector, sailing or even sawing. In the early 20th century, the usage of windmills became widespread as an off-grid solution for rural areas where electricity was not yet accessible[3].

However, the first significant step toward the use of wind for electricity generation came in the 1960s with experiments led by the Brace Research Institute at McGill University. This research aimed to provide renewable energy in remote communities, particularly in developing regions and for water management. By the mid-70s, due to the global oil crisis, wind energy gained more recognition in Canada, which encouraged research and development of alternative energy sources. It wasn't until the early 1990s that the commercialization of wind energy in Canada began. The first large-scale wind farm was established in Alberta, in 1994, marking the start of wind power's contribution to the national grid[3].

Today, wind energy plays an important role in Canada's net-zero emission target by 2050. It is in fact, the new cheapest source of electricity generation in the country. In 2022, the Canada had over 15,310 MW (megawatt) of installed wind power capacity, with around 337 wind energy projects across the country[4]. As shown in figure 1, wind energy contributes approximately 6% of Canada's total electricity generation, making it the second-largest source of renewable energy after hydroelectricity.

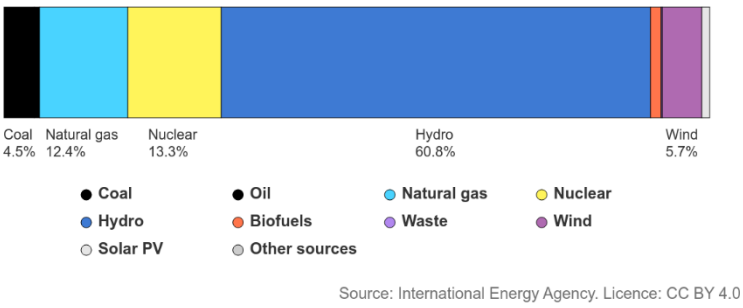

Figure 1: Electricity generation sources, Canada, 2022[5]

Canada has great potential of producing energy, wind resources are particularly abundant offshore and on the coastlines. However, these areas are far from the existing electrical grid. Which is why no offshore wind farm have been built yet. Wind farms are strategically located onshore in areas with consistent wind patterns, for example Alberta, Ontario, Quebec and the southern prairies[1]. Those farms produce around 36.06 TWh from wind energy[4]. Additionally, Technological advancements have improved turbine efficiency, allowing newer wind farms to generate more electricity with fewer turbines, further reducing the environmental footprint[6].

Table 1 presents the high concentration of wind installations in Ontario and Quebec, with well-developed wind farms[4]. However, provinces like Alberta and British Columbia also show considerable growth potential. Other regions like Manitoba still have a lot of room for expanding their wind capacity. The relationship between installed capacity and the number of wind farms highlights trends in project size and investment strategies across Canada. Moreover, bigger provinces with larger populations tend to possess greater installed capacities and a higher number of wind farms, showing a correlation between population size and the size of wind energy infrastructure.

Table 1: Wind Energy Capacity and Infrastructure by Province in Canada

Province	Number of inhabitants	Number of wind farms	Installed capacity (MW)
Ontario	15 996 989	91	5 000
Quebec	9 030 684	42	4 000
Alberta	4 849 906	45	3 000
British Columbia	5 646 467	8	600
Saskatchewan	1 231 043	10	300
Manitoba	1 484 135	2	250

The future of wind energy in Canada looks promising, with substantial growth expected over the coming decades. The government plans to expand the production of wind farms across the country and “proposed 600 million CAD (412.8 million EUR; 443.4 million USD) over seven years starting in 2022-2023 to Natural Resources Canada for the Smart Renewables and Electrification Pathways Program to support additional renewable electricity and grid modernisation projects.”[4]

Besides, several provinces have set ambitious renewable energy targets, and wind power is projected to play a key role in meeting these goals. Nova scotia, for example, plans on using offshore wind energy to produce hydrogen.[4]

Offshore wind farms provide also a promising future and can boost wind power capacity due to the stronger and more consistent wind speed. Although there is no example of these projects in Canada, offshore wind farms could offer a low cost and economically viable solution to energy production.[7]

To conclude, wind energy is experiencing significant growth, with new and more efficient installations across the country. The sector is supported by the government and private companies, while public opinion remains positive, recognizing wind energy as a crucial part of Canada's sustainable energy future[8].

References

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